

Hexuan (Matt) Wang

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Research Interests

My research lies in **Natural Language Processing and Multimodal Intelligence**, focusing on **faithful, grounded, verifiable, and reliable** generation and reasoning, such that model outputs can be traced to evidence and validated by humans. I investigate this across **attributed generation and multimodal grounding, temporal reasoning and generalization**, and **epistemic calibration**, using rigorous evaluation, post-training, and neuro-symbolic representations. Ultimately, I aim to develop trustworthy systems that augment human intellect in scientific discovery, complex decision-making, and collaborative learning.

Education

Johns Hopkins University

Aug 2023 – May 2027 (Expected)

B.S. in Computer Science, Applied Mathematics & Statistics

GPA: 3.98 / 4.00

- **Graduate-level Coursework:** Advanced Self-Supervised Statistical Models, Machine Social Intelligence, Machine Programming, Multimodal Understanding & Generation, Matrix Analysis.
- **Selected Coursework:** Natural Language Processing, Machine Learning, Probabilistic Models of the Visual Cortex, Algorithms, Optimization, Image Analysis, Numerical Linear Algebra, Statistics, Probability.

Publications & Preprints

- [1] **Hexuan Wang**, Y. Ren, S. Bommireddypalli, S. Chen, A. Prabhudesai, R. Zhou, E. Baral, P. Koehn. “SciTaRC: A Plan-Annotated Scientific Tabular QA Benchmark for Language Reasoning and Complex Computation.” *Accepted at COLM 2026* (early version at *SURGeLLM Workshop @ ACL 2026*). [arXiv:2603.08910](#).
- [2] **Hexuan Wang**, J. Zhang, B. Van Durme, D. Khashabi. “Are Finer Citations Always Better? Rethinking Granularity for Attributed Generation.” *Under review at TMLR*. [arXiv:2604.01432](#).
- [3] **Hexuan Wang**, M. Brownrigg, C. Rispoli, T. Li, J. Zhang, X. Wang, B. Van Durme, F. Faghri, M. Farajtabar, E. Nalisnick, D. Khashabi. “Training in the Past, Reasoning about the Future: Principled Evaluation of Temporal Generalization.” *Preprint and open-source release planned for October 2026*.

Research Experience

Faithful LLM Reasoning: Evaluation, Verification & Training

May 2025 – Present

Advisor: Prof. Daniel Khashabi (Computer Science, CLSP)

Johns Hopkins University

- **Chrononauts.** Trained and evaluated 7 temporal cutoff models to disentangle what models inherently know from how well they reason forward in time. Built retrieval systems over Wikipedia snapshots, curated custom training data, and extended a 3B model to 8k context. Developed custom agentic post-training (SFT and RL) for interleaved search and reasoning, outperforming larger baselines (trained on far more compute) on multi-hop reasoning while demonstrating robust temporal generalization and calibration. *Outcome:* first-author paper [3] and open-source release planned for October 2026.
- **Citation Granularity.** Evaluated citation granularity in attributed generation across 4 model scales (8B–120B), controlling for citation volume to isolate segmentation effects. Discovered that enforcing fine-grained (sentence-level) citations forfeits a median 40% in attribution quality by breaking semantic context, whereas intermediate (paragraph-level) granularity maximizes attribution with negligible impact on correctness. Showed that fine-grained constraints disproportionately penalize larger models and can invert benchmark rankings. *Outcome:* first-author paper [2], under review at TMLR, [arXiv:2604.01432](#).
- **Epistemic Awareness & Hallucination.** Investigated why reasoning models confidently hallucinate on logically underspecified problems instead of abstaining. Uncovered a stark representation–verbalization gap across open models (1.5B–70B): internal states reliably encode problem unsolvability, yet this signal fails to surface in generation. Demonstrated via causal interventions that internal recognition remains computationally inert at the answer step, explaining why models fail to self-correct or abstain. *Outcome:* supported by the Pistrutto Research Fellowship.

Interactive Figure-to-SVG Editing via Element-Level Grounding

Advisor: Prof. Jaemin Cho (Computer Science)

Mar 2026 – Present
Johns Hopkins University

- Leading the development of an interactive pipeline to convert scientific figures into editable SVGs, addressing both visual reconstruction and code-level editing. Built the underlying data engine and post-trained a 4B multimodal model via SFT and RL with element-level feedback, grounding code in rendered regions so agent workflows can resolve visual references (such as clicks) and execute localized edits. *Outcome*: lead-author project, ongoing.

Scientific Table Reasoning: Decoupling Planning from Execution

Advisor: Prof. Philipp Koehn (Computer Science, CLSP)

Dec 2024 – Mar 2026
Johns Hopkins University

- SciTaRC**. Conceived and led a team of 7 student collaborators to construct an expert-authored benchmark evaluating composite reasoning over scientific tables. Designed the diagnostic framework pairing questions with structured plans and complexity metrics to decouple strategy formulation from execution, coordinating dataset curation across the group. Demonstrated that providing oracle step-by-step plans yields limited gains, exposing a fundamental execution bottleneck in long-horizon computation over structured data. *Outcome*: first-author paper [1], accepted at COLM 2026, [arXiv:2603.08910](https://arxiv.org/abs/2603.08910).

Baseball Scheduling Optimization

Advisor: Prof. Anton Dahbura (Computer Science, ISI)

Jan 2024 – Feb 2025
Johns Hopkins University

- Formulated multi-objective mixed-integer programs (Gurobi, Hexaly) balancing travel, equity, rest, and commercial franchise constraints. *Outcome*: Adopted in production across Minor League Baseball (MiLB).

Spectral Analysis & Reconstruction of FM Synthesis

Advisor: Prof. Daniel Naiman (Applied Mathematics & Statistics)

Nov 2023 – Sep 2024
Johns Hopkins University

- Recreated DX7 FM synthesis in PyTorch; built an STFT algorithm tracking time-varying peaks to reconstruct audio. *Outcome*: Talk at MATRX, poster at SIAM DMV; [featured by JHU AMS News](#).

Honors & Awards

- Pistritto Research Fellowship** (\$6,000), Dept. of Computer Science, JHU. Competitive, faculty-selected fellowship supporting independent research in representation analysis and visualization. 2026
- Naddor Prize**, Dept. of Applied Mathematics & Statistics, JHU. Awarded for distinguished academic performance by a non-senior student in applied mathematics and statistics. 2026
- Dean's List**, Johns Hopkins University Every semester

Teaching

Teaching Assistant

Johns Hopkins University

Jan 2024 – Present
Baltimore, MD

- Computer Science**: NLP: Self-Supervised Models (601.471/671, F26), Artificial Intelligence (601.664, 3 terms: S25, F25, S26), Math Foundations for CS (601.230, F24).
- Applied Math & Statistics**: Numerical Linear Algebra (553.480/680, F25), Linear Algebra & Data Science (553.291/295, 3 terms: S24, S25, F25).

Technical Skills

- LLM/MLLM Training & Systems**: PyTorch, HuggingFace (Transformers, TRL, PEFT), OpenRLHF, DeepSpeed, FSDP, vLLM, Accelerate, continual pretraining, SFT, RL (RLHF/RLVR, DPO), LoRA.
- Evaluation, Retrieval & Probing**: lm-evaluation-harness, Weights & Biases (W&B), FAISS, BM25, linear probing (scikit-learn), causal intervention, HuggingFace Datasets.
- Optimization & Scientific Computing**: Integer programming (Gurobi, Hexaly), combinatorial optimization, Short-Time Fourier Transform (STFT), NumPy, SciPy, MATLAB, Mathematica.
- Languages & HPC Infrastructure**: Python, C/C++, Bash/Linux, Slurm, Singularity, Java, Git, $\LaTeX$.

Academic Service & Leadership

- Conference Reviewing**: NeurIPS 2026.
- JHU Vocal Chords A Cappella** — Business Manager & Bass Singer 2023 – Present
3rd Place, ICCA Quarterfinals (2024, 2026); managed \$3,000+ production and equipment budget.